CHARAN MADALA

VLSI PHYSICAL DESIGN ENGINEER

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PROFESSIONAL SUMMARY

Electronics and Communication graduate building a career in VLSI Physical Design, with a strong foundation in CMOS, digital design, and Verilog HDL and hands-on RTL-to-GDS experience using Cadence Genus and Innovus. Practical exposure to synthesis, floorplanning, macro placement, power planning, placement, and clock tree synthesis, gained on a CVA6 RISC-V processor (approx. 100K instances, 24 macros), an AES cryptographic core, and multi-clock synthesis test cases. Skilled in Tcl scripting and Linux for flow automation and report analysis.

Career Goal: To grow as a Physical Design Engineer by contributing to block- and chip-level implementation, and to deepen my expertise in timing optimization and timing closure on advanced, high-performance designs, progressing toward sign-off and ownership of complete physical design flows.

TECHNICAL SKILLS

EDA Tools: Cadence Genus, Cadence Innovus | **HDL:** Verilog | **Scripting / OS:** Tcl, Linux
Physical Design: Synthesis, Floorplanning, Macro Placement, Power Planning, Placement, CTS
Fundamentals: CMOS, Digital Design, Combinational and Sequential Logic
Design Concepts: Clock Domains, Generated Clocks, Standard Cells, Timing Constraints

VLSI TRAINING

Physical Design Training | Sumedha IT June 2026 – December 2026

- RTL-to-GDS implementation using Cadence Genus and Innovus, covering floorplanning, macro placement, power planning, placement, and clock tree synthesis.
- Practicing Tcl scripting and Linux commands for design automation and implementation analysis.

VLSI PROJECTS

CVA6 RISC-V Processor | Physical Design *CTS In Progress*

Tools: Cadence Genus, Cadence Innovus / Target: 1.5 ns clock period

- Implementing the CVA6 RISC-V processor through the RTL-to-GDS flow; completed synthesis and standard-cell placement, and currently progressing through clock tree synthesis.
- Handling a large design of approx. 100K instances and 24 macros, which requires careful macro placement and power planning.
- Applied timing constraints for a single clock domain with a 1.5 ns target clock period, and analyzed implementation results at each stage.
- Using Tcl scripts and Linux commands to automate tool runs and review implementation reports.

AES Core | Cryptographic Engine

Tools: Cadence Genus, Cadence Innovus / Target: 1 ns clock period

- Performed synthesis and placement of an AES-based cryptographic hardware block with approx. 10K standard cells and a single clock domain.
- Implemented the design against an aggressive 1 ns target clock period, applying timing constraints during synthesis.
- Gained practical exposure to the physical implementation of a cryptographic block and the effect of tight timing targets on the flow.

Multi-Clock Design | Synthesis Test Cases

Tools: Cadence Genus / Scope: 3 clock domains, approx. 1K standard cells

- Synthesized test designs containing three clock domains, including a generated clock.
- Practiced writing multiple clock definitions and generated-clock constraints and verifying them during synthesis.
- Built a clear understanding of how clock definitions in the constraints affect timing analysis across domains.

EDUCATION

M.Sc. Computing and Technology Jan 2022 – Jan 2024
Northumbria University Result: 61/100

B.Tech. Electronics and Communication Engineering Jun 2017 – May 2021
Sathyabama University, Chennai CGPA: 7.3/10

CERTIFICATIONS

NPTEL – Data Structures and Computing with Python
Coursera – Introduction to VLSI Flow

LANGUAGES

English – Professional Working Proficiency
Telugu – Native
Hindi – Conversational